Hardware-Aware AI-Assisted Beam Training, Tracking, and Channel Estimation for RIS-Assisted mmWave UAV Links

Maavia Khalid and Hassan Ali
Department of Computer Science
Institute of Space Technology (IST)
Islamabad, Pakistan

***Abstract*—Millimetre-wave (mmWave) and sub-terahertz links can give unmanned aerial vehicles (UAVs) multi-gigabit connectivity, but narrow beams, high path loss and UAV mobility make beam management costly. We present a hardware-aware framework for a 60 GHz UAV link assisted by a reconfigurable intelligent surface (RIS) that models transmitter power-amplifier distortion, RIS phase quantization, phase noise, amplitude–phase coupling and antenna mismatch, together with ITU-R P.676-13 gaseous absorption. Impairments cost 7.8 dB of SNDR, mostly from amplitude–phase coupling, so a realistic RIS needs about 2.25× more elements than an ideal one. A position-parametric channel estimator comes within 0.15 dB of perfect channel knowledge with 64× fewer pilots than per-element LMMSE, and predictive tracking keeps 99.3% of the ideal array gain at 500 ms latency, against 63% for persistence. On real 60 GHz measurements (DeepSense 6G Scenario 23), we show that exact blind beam prediction is bounded at 79% top-1 by measurement noise, and propose AI-assisted reduced beam training with a learned beam-tie graph. In a closed-loop replay of 33 flights, where the system knows only the beams it measured, candidate beams from the UAV's GPS and from its position in the base station's camera image, located without labels, let the tracker select the exact optimal beam 99.5% of the time, with macro-F1 0.97 over all 56 beam classes and 0.66 ms of beam training per decision instead of 5 ms. Ten deep sequence models do not beat the tie graph at equal measurement budget. All results use sequence-level cross-validation and per-flight significance tests.**

***Index Terms*—mmWave, sub-THz, UAV, reconfigurable intelligent surface, beam training, beam tracking, channel estimation, hardware impairments, deep learning.**

I. Introduction

Sixth-generation (6G) systems target mmWave and sub-THz bands for their large bandwidths [1], [2]. UAVs benefit directly, as aerial base stations, relays and sensing platforms, but high-frequency links depend on narrow beams that a moving UAV quickly leaves. Exhaustive beam sweeping costs time that a fast link cannot spare, which has motivated learning-based beam prediction from position and vision [3], [4]. Reconfigurable intelligent surfaces (RISs) add a programmable reflection path when the direct link is blocked [14], but real RIS hardware and transceivers are not ideal: transceiver distortion and RIS phase noise change both performance and the optimal configuration [5], [6], and a practical RIS element couples its reflection amplitude to its phase [15].

Three gaps motivate this work. First, RIS–UAV studies often pair idealised hardware with idealised link budgets, which hides the hardware cost. Second, phase-tracking studies frequently score controllers by per-element phase error, a metric that, as we show, says nothing about received power. Third, beam prediction results on public datasets are easy to overstate: in the dataset we use, the ground-truth beam is the argmax of the same power sweep often used as input, and consecutive samples are nearly identical, so random splits leak.

Contributions.

1) A hardware-aware RIS–UAV link model with transmitter PA distortion (Bussgang SNDR), RIS phase quantization, phase noise, amplitude–phase coupling and S11 mismatch, on a full ITU-R P.676-13 absorption model, with an impairment ablation and an RIS sizing rule (Sec. II, VI-A).
2) Hardware-aware control and estimation: a coupling-aware discrete phase search, an LMMSE estimator that inverts the hardware's realised training patterns, and a position-parametric estimator with an adaptive pilot budget (Sec. III, VI-B).
3) A predictive tracking study scored by delivered array gain, showing that position-based RIS control works at realistic sensor precision once the right metric is used (Sec. IV, VI-C).
4) On real 60 GHz measurements, a proof that exact blind beam prediction is noise-limited, a GRU trained on a soft power target, a learned beam-tie graph, and an AI-assisted reduced beam training scheme meeting ≥ 95% exact top-1 and macro-F1, conformal shortlists calibrated to a target accuracy, a closed-loop replay of every multi-sweep flight in which the system knows only what it measured, and a matched-budget comparison of ten deep sequence models against the tie graph in that loop, evaluated with sequence-level cross-validation and bootstrap confidence intervals (Sec. V, VI-D).
5) A label-free UAV locator for the base station's camera and a sensing-aided tracker that uses camera and GPS candidates instead of periodic sweeps, halving beam-training latency while raising exact top-1 above 99% and all-class macro-F1 above 0.95, robust to stale GPS and to the assumed timing, and a nested stack of blind

predictors that reaches 96% of the noise bound, with per-flight significance tests for every comparison (Sec. VI-D).

II. System Model

Fig. 1 summarises the two parts of this work: a hardware-aware model of the RIS-assisted link (Secs. II–IV), evaluated in simulation, and beam management on real 60 GHz measurements (Secs. V–VI), where sensing proposes candidate beams that are then measured.

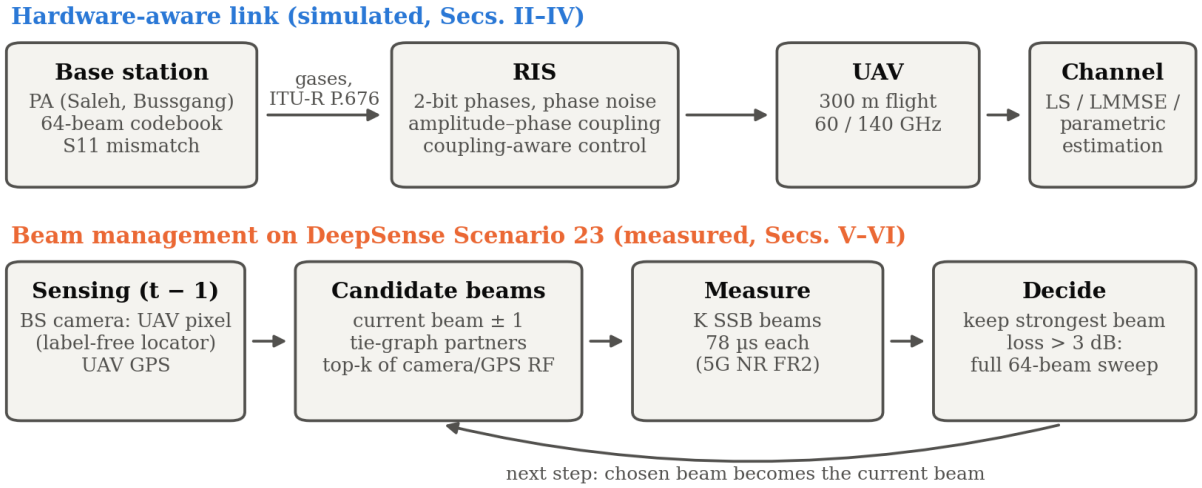


Fig. 1. System overview. Top: the simulated hardware-aware link, from the BS power amplifier through the impaired RIS to the UAV, and channel estimation. Bottom: the sensing-aided beam-management loop evaluated on DeepSense Scenario 23; the camera and GPS of the previous step propose candidates, the base station measures them with SSB beams, keeps the strongest, and sweeps all 64 beams only on beam loss.

A. Geometry

A fixed base station (BS) at (25, −15, 10) m serves a UAV through an N-element RIS whose direct path is blocked. The RIS is a 32×32 uniform planar array with λ/2 spacing (8 cm aperture at 60 GHz), mounted on a facade at (0, 0, 20) m with outward normal +x; it serves only the half-space in front of it. The UAV flies at 10 m/s; trajectories cover straight, diagonal, climbing, descending, circular, helical and waypoint paths, all in front of the panel.

B. Channel

For element n at distance d, the BS→RIS and RIS→UAV coefficients are

$$h_1[n] = \sqrt{G_{BS}} \cdot (\lambda / 4\pi d_{1n}) \cdot e^{-j2\pi d_{1n}/\lambda} \cdot a(d_{1n}) \cdot \sqrt{(G_e \cos^q \theta_{1n})}, \quad (1)$$

and likewise $h_2[n]$ with the UAV gain G_{UAV}, where $a(d) = 10^{-\gamma d/20}$ is the gaseous-absorption amplitude with γ from the ITU-R P.676-13 line-by-line model (44 O_2 and 35 H_2O lines) at 1013.25 hPa, 288.15 K and 7.5 g/m³ [8]; γ = 14.6 dB/km at 60 GHz. $G_e = \pi$ is the gain of a $(\lambda/2)^2$ element [16], θ is measured from the panel normal, q = 1.5, and the pattern is zero behind the panel. G_{BS} = 25 dBi and G_{UAV} = 10 dBi; the transmit power is 20 dBm and the receiver noise is kTB·NF = −87 dBm (B = 100 MHz, NF = 7 dB).

The received signal with reflection coefficients Γ_n is

$$y = \Sigma_n h_2[n] \Gamma_n h_1[n] \cdot x + w. \quad (2)$$

With ideal hardware the optimal phases are $\varphi_n = -\angle(h_1[n] h_2[n])$ and the array gain grows as N^2 (verified: +12.04 dB per quadrupling of N).

C. Hardware Impairments

Transmitter PA. A passive RIS has no amplifier; the PA sits at the BS. We apply Saleh's AM/AM and AM/PM model [9] ($\alpha = 2.1587$, $\beta = 1.1517$, $\alpha_\varphi = 4.0033$, $\beta_\varphi = 9.1040$) to a complex Gaussian (OFDM-like) input at 6 dB input back-off. By the Bussgang decomposition [17] the output is K·x + d, with d uncorrelated with x, so a fraction ρ_d of the output power is distortion that travels through the channel with the signal [18]:

$$\mathrm{SNDR} = \rho_u P|h|^2 / (\rho_d P|h|^2 + \sigma^2). \quad (3)$$

At 6 dB back-off ρ_u/ρ_d gives an SNDR ceiling of 13.3 dB, independent of link strength.

RIS. Commanded phases are quantized to b = 2 bits, perturbed by Gaussian phase noise (σ = 5°), and realised with the phase-dependent amplitude of a practical element [15]:

$$\beta(\varphi) = (1 - \beta_{min}) \cdot ((\sin(\varphi - \phi) + 1)/2)^\kappa + \beta_{min}, \quad (4)$$
$$\beta_{min} = 0.2,\ \phi = 0.43\pi,\ \kappa = 1.6,$$

scaled by a reflection efficiency of 0.85.

Antenna mismatch. S11 = −10 dB at the BS and UAV ports gives a power transfer of $1 - |S11|^2$ each.

Each impairment is switchable, so ideal and hardware-aware links are compared on identical geometry.

III. Hardware-Aware Control and Channel Estimation

A. Coupling-Aware Phase Selection

Because (4) ties amplitude to phase, the ideal phases are no longer optimal. For a common output phase ψ the best realisable level for element n maximises $\mathrm{Re}(c_n\,\Gamma(\text{level})\,e^{-j\psi})$, with $c_n = h_1[n]h_2[n]$; we sweep 64 values of ψ and keep the best. The search is fully vectorised (≈3 ms for N = 1024).

B. Per-Element Estimation: LS vs Hardware-Aware LMMSE

The RIS cycles through N Hadamard training patterns, each held for L pilot symbols. The *naive* LS estimator assumes the commanded $0/\pi$ states were realised exactly. Under (4) the two states have amplitudes 0.19 and 0.91, which biases LS. The *hardware-aware* estimator trains on the antipodal state pair with the largest reflected energy, models the realised coefficients through (4), and applies LMMSE with an i.i.d. prior; because PA distortion is signal-dependent, only its average power enters the noise model.

C. Position-Parametric Estimation

The BS and RIS are fixed, so $c(p) = h_1 \odot h_2(p)$ depends only on the UAV position p. Beyond the Fresnel distance (≈5 m here) the pilots constrain the direction of p sharply and its range weakly. Starting from a prior p_0, the RIS transmits M random realisable patterns A, and the receiver estimates the direction by maximising the likelihood under an unknown complex gain,

$$\hat{p} = \mathrm{argmax}_p\, |y^H A\, c(p)|^2 / \|A\, c(p)\|^2, \qquad (5)$$

over direction cosines on a coarse grid finer than the beam width, then a fine grid. The RIS is then configured from $c(\hat{p})$ with the search of Sec. III-A.

Adaptive pilot budget. The search succeeds when the matched-filter pilot SNR $M \cdot L \cdot \mathrm{SNR}_{coh}/N$ exceeds a threshold; SNR_{coh} is predicted from p_0 and the known geometry. With M = 64 we choose L to reach a threshold of 128, selected on random seeds separate from those reported. If the prior's angular uncertainty is already below one eighth of the beam width, no pilots are sent.

IV. Predictive RIS Tracking

The RIS must be configured for where the UAV will be once measurement, computation and control latency have passed. We predict the position h steps ahead ($h \cdot 10$ ms, $h \in \{1, 10, 50\}$) from RTK-class position measurements ($\sigma = 2$ cm per axis) with four predictors: persistence, constant-velocity extrapolation, a constant-velocity Kalman filter [13] whose acceleration PSD is tuned on training trajectories, and a ridge regressor on an 8-sample window trained on 60 random trajectories. All predictors see the same noisy measurements, and the nine designed trajectories are held out.

Metric. Mean absolute per-element phase error counts a phase offset common to all elements as error, but such an offset leaves $|\Sigma_n\, c_n\, e^{j\hat{\varphi}_n}|$ unchanged. We therefore score the fraction of the ideal coherent gain delivered at the true position:

$$G = |\Sigma_n\, c_n\, e^{j\hat{\varphi}_n}|^2 / (\Sigma_n\, |c_n|)^2. \qquad (6)$$

V. AI Beam Prediction on Real Measurements

A. Dataset and Evaluation Protocol

DeepSense 6G Scenario 23 [10] provides 11,387 samples of 64-beam received-power sweeps at 60 GHz from a drone flight campaign, organised into 35 flight sequences, with the optimal-beam index as label (56 classes in use). Three properties of the data shape the protocol:

1) **The label is the argmax of the sweep** for all 11,387 samples, so predicting the current beam from the current sweep is trivial (100% by argmax). Every task below predicts the beam at time t + h from sweeps up to time t.
2) **Consecutive samples are near-duplicates** (≈7 Hz). All models are evaluated with GroupKFold(5) over flight sequences, so a test sequence is never seen in training.
3) **Near-ties are noise.** In 75.3% of sweeps the best and second-best beams are within 0.25 dB (they are adjacent in 90.6%), while the same beam's power changes by a median 0.18 dB between consecutive sweeps.

Comparisons between two methods are tested per flight, the unit of replication, because samples within a flight are strongly correlated: we report the pooled difference with a 95% CI from a paired bootstrap over flights (5000 resamples) and the Wilcoxon signed-rank test on per-flight differences, Holm-corrected within each family of comparisons.

B. Blind Prediction

A two-layer GRU (hidden size 128) reads the last 8 sweeps (dB relative to peak) and the sampling gaps, and outputs beam logits added to a learned multiple of the last sweep, so that it learns a residual over persistence. Because near-tied labels are noise, we train against a **soft target**, $\mathrm{softmax}(P_{t+h}/0.5\ \mathrm{dB})$, the measured power of the future sweep, rather than its argmax; a hard-label GRU is the ablation. Early stopping uses held-out training sequences. Our final predictor averages the log-probabilities of three such GRUs (different seeds) trained with **beam-axis augmentation**: each batch is shifted by up to ±3 beams and mirrored with probability 0.5, identically for history, last sweep and target, with vacated edge beams set to the row's floor rather than wrapped. A UAV at a slightly different angle sees a shifted beam pattern, so this adds plausible examples of rarely visited beams. Random-forest and extra-trees baselines use the last 3 sweeps.

Stacked predictor. Different predictors make different errors, so our strongest blind predictor combines them. Each component's scores are standardised per sample and summed with weights chosen on the training folds by coordinate ascent (objective: within-noise accuracy plus half the exact top-1) and applied unchanged to the held-out fold. The components are the GRU variants trained with each set of side information (Sec. VI-D), extra trees and LightGBM [25] on long-history features, the last sweep, and a shifted-template tracker that follows the UAV's sub-beam peak position across sweeps.

Noise ceiling. As an upper bound for any causal predictor of the exact label, a non-causal oracle that averages the sweeps

before and after the target sweep reaches only **79.0% exact top-1 and 0.59 macro-F1**.

C. AI-Assisted Reduced Beam Training

A deployed link does not have to choose blind: it can measure a few candidate beams at time t + h and keep the strongest. The predictor ranks the beams; the system measures the top K and keeps the best; the result is scored against the **full 64-beam ground truth** on exact top-1 and macro-F1, with an overhead of K/64 of a full sweep. Shortlists are produced from out-of-fold GRU scores and compared with non-learned rankings (the last sweep's strongest beams; a window around the last best beam). The fused ranking adds the z-scored GRU log-probabilities to the z-scored last sweep with equal weight. An adaptive variant measures the smallest set whose GRU probability reaches τ.

Learned tie graph. The strong beams of this codebook are not contiguous: beams 27, 28, 32, 33 and 38 are often within 0.1 dB of each other, so a search around the last best beam misses the true best. For each beam b, the training flights give the beams most often within 1 dB of the best when b was the best. Two schemes use it, following the coarse/refine split of 3GPP beam management. *Two-round*: measure the first K_1 ranked beams, then the M top graph partners of the measured winner; if the best measured power is still more than 3 dB below the last sweep's peak, the beam is treated as lost and the remaining beams are swept. *Burst*: append the graph partners of the two *predicted* best beams to the first K ranked beams and measure them all at once, avoiding a report round trip.

Latency model. We use 5G NR FR2 timing at 120 kHz subcarrier spacing, where up to 64 SSB beams fit in a 5 ms half-frame [22]: 78 µs per beam measurement, plus 1 ms for each extra report round trip. A full sweep costs 5 ms. GRU inference (0.5 ms for one model and 1.4 ms for the 3-model ensemble at batch size 1 on a laptop CPU) is not on the critical path: the shortlist for t + h is computed from sweep t, during the ≈140 ms before t + h.

Operating points by nested cross-validation. For each held-out fold, the setting (K, τ, or K_1, M and the fallback) is the lowest-latency one meeting 95% on both exact top-1 and all-class macro-F1 over the other four folds, and is then applied unchanged to the held-out fold. The tie graph is always fitted on folds other than those it is scored on, including inside this selection. 95% confidence intervals come from 500 bootstrap resamples of whole flight sequences.

Conformal shortlists. A measured set returns the exact optimal beam if and only if that beam is in the set, so set coverage equals exact top-1. We calibrate the set size by split conformal prediction [23]: on four folds, the score is the true beam's rank in the GRU ranking, and the threshold is its $\lceil (n+1)(1-\alpha) \rceil$-th smallest value; the fifth fold is the test. Adaptive prediction sets [24] use the GRU's cumulative probability instead. Samples within a flight are correlated and flights differ, so the exchangeability behind the guarantee holds only approximately; we report the coverage of every held-out fold.

Closed-loop replay. The schemes above build each shortlist from full sweeps of earlier steps, which a system measuring only K beams per step never has. We therefore replay every flight with more than one sweep (33 of 35) step by step with policies that know only the beams they measured: the current beam, its ±1 index neighbours and its M strongest tie-graph partners (fitted on other folds' flights), with a full sweep every N steps or only on beam loss (best measured power more than 3 dB below the current beam's last value). The effective spectral efficiency over each interval is $(1 - T_{train}/T)$ times the mean of $\log_2(1 + SNR_0\, p_{used}/p_{best})$ at the interval's start and end, using measured powers, so beam ageing is measured, not modelled. The dataset's powers are unitless, so SNR_0, the best beam's SNR, is a nominal operating point.

Learned closed-loop tracking. To test whether a learned model helps once only measured beams are known, we log each flight under the tie-graph tracker (±1 window, four partners, full sweep every 8 steps). Each logged step holds the measured beams' power in dB relative to the strongest measured beam, −20 dB for unmeasured beams, the measurement mask, a full-sweep flag and the time gap. A model reads the last W steps and predicts the next step's best beam; the training target is the next recorded full sweep (soft, 0.5 dB temperature, or hard), available offline only. In deployment the tracker measures its usual set plus the model's top $k \in \{2, 4\}$ beams not already in it, and the model's next input is built from what was measured. The baseline at the same budget is the tracker with k more tie partners. We compare a GRU, an LSTM, a dilated temporal CNN, a Transformer encoder and an MLP over the window, and GRU variants in hidden size (64, 128), learning rate (10^{-3}, $3\cdot10^{-4}$), window (4, 8, 16) and target (soft, hard); all are trained with AdamW, batch 128, early stopping on validation flights. The headline configuration is selected by top-4 accuracy on validation flights inside the training folds. Inference latency is the median of 300 batch-1 forward passes on one thread of a laptop CPU, in fp32 and with dynamic int8 quantization; it is not a measurement on embedded hardware.

VI. Results

A. Link Budget and Hardware Impairments

With ideal hardware the SNR is 15.8 dB at 20 m and 8.8 dB at 50 m. At 50 m the combined impairments cost **7.83 dB** (Fig. 2): amplitude–phase coupling 5.51 dB, PA nonlinearity 1.46 dB, S11 mismatch 0.92 dB, 2-bit quantization 0.91 dB and phase noise 0.03 dB.

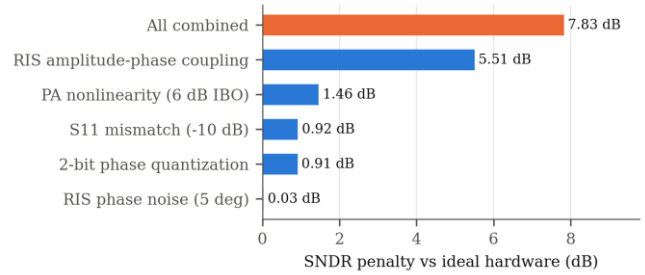


Fig. 2. SNDR penalty of each impairment alone and combined, UAV at 50 m.

Note on the first draft. Without antenna and element gains and with an 8×8 (2 cm) RIS, the SNR is −43.2 dB at 20 m, the value reported in our earlier draft; it reflects missing gains, not sub-THz propagation. The same cause explains the earlier −170 dBm at 140 GHz: with gains included, an 8×8 RIS receives −115.9 dBm at 140 GHz at 50 m, and the N^2 law holds at both carriers (Table I).

TABLE I
RECEIVED POWER AT 50 M, IDEAL HARDWARE (DBM)

RIS	60 GHz	140 GHz
8×8	−102.3	−115.9
16×16	−90.2	−103.8
32×32	−78.2	−91.8
64×64	−66.2	−79.7

Phase resolution and coupling strength. Coupling-aware selection (Sec. III-A) gains more as more realisable states become available: 0.55, 0.92 and 1.01 dB at 2, 3 and 4 bits, and nothing at 1 bit (−0.06 dB). No measured coupling curve for a 60 GHz element is available to us, so we sweep its strength (Table II). The total penalty ranges from 2.9 dB without coupling to 7.8 dB at β_{min} = 0.2; the conclusions below hold across this range, and the gain of coupling-aware control shrinks as coupling weakens.

TABLE II
SNDR PENALTY VS COUPLING STRENGTH, UAV AT 50 M, 2-BIT PHASES

β_{min}	0.2	0.4	0.6	0.8	1.0 (none)
Naive control (dB)	7.83	6.22	4.92	3.84	2.93
Coupling-aware (dB)	7.28	5.98	4.83	3.80	2.90

Impairments change how large the RIS must be (Fig. 3). Over a 20 → 320 m flight, a 64×64 RIS gives 100% availability (SNDR ≥ 0 dB) with ideal hardware but only 57% with realistic hardware; a 96×96 RIS (24 cm) restores 100%. The hardware cost is therefore about 2.25× in element count.

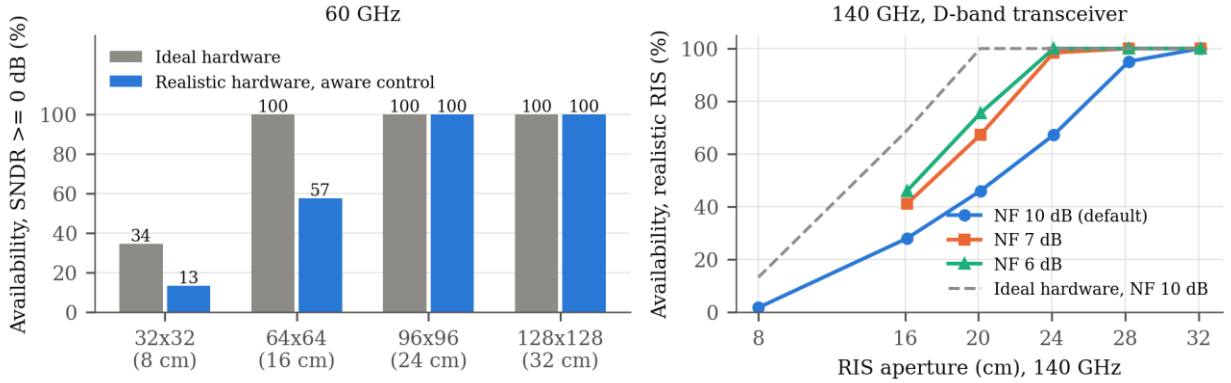


Fig. 3. Link availability over the flight versus RIS size. Left: 60 GHz, ideal versus realistic hardware with coupling-aware control. Right: 140 GHz with D-band transceiver hardware, realistic RIS, for receiver noise figures of 6, 7 and 10 dB (dashed: ideal hardware).

60 vs 140 GHz at equal aperture. With the same 8 cm panel (32×32 at 60 GHz, 75×75 at 140 GHz) and the same transceiver hardware, 140 GHz performs slightly better (ideal SNR 16.6 vs 15.8 dB at 20 m; 47.5% vs 34.4% availability): it avoids the 60 GHz oxygen line (0.69 vs 14.6 dB/km) and fits 5.5× more elements in the same area. With published D-band hardware (about 20 dBm PA saturation [20], hence 14 dBm average at 6 dB back-off) and a conservative 10 dB receiver noise figure the link loses about 9 dB, and the RIS must grow to 32 cm (300×300 elements) for full availability with realistic impairments, against 24 cm at 60 GHz (Table III). The receiver noise figure is the decisive assumption: D-band SiGe LNAs alone measure 4.8–6.1 dB [21], and with a 6 dB receiver 24 cm suffices at 140 GHz as well (7 dB: 28 cm). At sub-THz, transceiver hardware, not propagation, sets the RIS size, and the panel spans 24–32 cm across that range.

Sensitivity to assumed values. Three further inputs are literature values or assumptions rather than measurements, so we vary each. (i) The Saleh coefficients are Saleh's travelling-wave-tube fit; how much they matter depends on how hard the PA is driven. From 3 to 12 dB input back-off the PA alone costs 2.6 to 0.4 dB at 50 m, and all impairments with coupling-aware control 7.8 to 6.9 dB, so amplitude–phase coupling remains the dominant impairment at every operating point. (ii) For element obliquity exponents from 0 (isotropic) to 2, the 60 GHz RIS needs 64×64 elements for full coverage with ideal hardware and 96×96 (24 cm) with realistic hardware, the same 2.25× as with the default 1.5; only a strongly directive element (exponent 3) changes it, to 80×80 and 112×112 (1.96×). (iii) Around the assumed 14 dBm D-band average power, 11 and 17 dBm move the 140 GHz panel to 36 and 28 cm (NF 10 dB).

TABLE III
AVAILABILITY (%) OVER THE FLIGHT AT 140 GHZ WITH D-BAND HARDWARE (14 DBM AVERAGE TRANSMIT POWER), BY RIS SIZE AND RECEIVER NOISE FIGURE (NF)

RIS (aperture)	75×75 (8 cm)	150×150 (16 cm)	188×188 (20 cm)	225×225 (24 cm)	263×263 (28 cm)	300×300 (32 cm)
Ideal hardware, NF 10 dB	13.1	68.9	100	100	100	100
Realistic, coupling-aware, NF 10 dB	1.6	27.9	45.9	67.2	95.1	100
Realistic, coupling-aware, NF 7 dB	–	41.0	67.2	98.4	100	100
Realistic, coupling-aware, NF 6 dB	–	45.9	75.4	100	100	100

B. Channel Estimation

Naive LS saturates at about −7.5 dB NMSE because of the coupling bias, whereas hardware-aware LMMSE keeps improving with pilot length (−11.1 dB at 20 m with 4096 symbols per pattern) and is never worse. In closed loop (estimate, configure, measure SNDR), the hardware-aware pipeline beats the naive one at every distance and pilot length, by 0.4–1.2 dB.

Per-element estimation, however, scales poorly with range. The parametric estimator (Table IV) reaches perfect-CSI SNDR to within 0.15 dB at 20–100 m with 64×1024 = 65k pilot symbols, against 4.2M for per-element LMMSE, which still falls 1.6 dB short at 100 m. A 5 m GPS prior alone is useless (−23 to −31 dB) because its error exceeds the beam width. With the adaptive budget (Table V), the estimator delivers 96.5–100% of perfect-CSI SNR on average at every range and prior, and needs no pilots when the 100 ms Kalman prediction is used as prior.

TABLE IV
SNDR (DB) FROM A 5 M GPS PRIOR

UAV at	Perfect CSI	GPS only	Parametric, 65k pilots	Per-element LMMSE, 4.2M pilots
20 m	7.47	−23.7	7.47	7.32
50 m	1.50	−31.4	1.49	1.07
100 m	−4.75	−22.9	−4.90	−6.33

TABLE V
ADAPTIVE PARAMETRIC ESTIMATION: MEAN (WORST OF 10) % OF PERFECT-CSI SNR, AND PILOT SYMBOLS

UAV at	GPS prior (5 m)	Kalman, 500 ms (31 cm)	Kalman, 100 ms (5 cm)
20 m	97.4 (90.9), 6k	97.3 (93.4), 4k	99.5 (98.6), 0
50 m	98.4 (95.9), 31k	98.5 (95.8), 33k	99.9 (99.4), 0
100 m	96.9 (93.5), 131k	96.8 (91.7), 131k	99.9 (99.6), 0
200 m	96.5 (86.3), 524k	99.4 (98.5), 0	100.0 (99.8), 0

C. Predictive Tracking

Every predictor shows a mean absolute phase error of 89–92°, the value an unaligned phase profile would give, yet the delivered gain (6) ranges from 63% to 100% (Table VI, Fig. 4). The phase-error metric therefore cannot rank controllers. Prediction matters at realistic latencies: at 500 ms the Kalman filter delivers 99.3% of the ideal gain (5th percentile 97.4%), the learned ridge predictor 98.8%, constant velocity 71.4% and persistence 63.0%. All predictors run in under 0.1 ms per step.

TABLE VI
DELIVERED ARRAY GAIN (%) AND POSITION RMSE (CM), HELD-OUT TRAJECTORIES

Horizon	Persistence	Constant velocity	Kalman CV	Ridge (learned)
10 ms	99.9 / 10.4	99.9 / 7.7	**100.0 / 2.0**	100.0 / 2.7
100 ms	94.0 / 97.9	97.2 / 51.9	**99.98 / 5.2**	99.9 / 8.2
500 ms	63.0 / 488	71.4 / 250	**99.3 / 30.7**	98.8 / 41.4

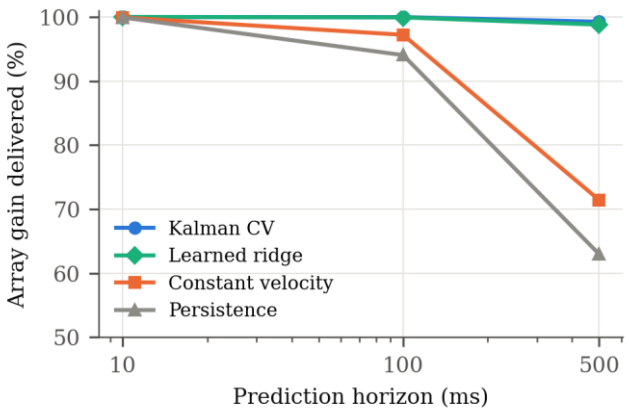


Fig. 4. Delivered array gain versus prediction horizon (held-out trajectories); the Kalman and the learned ridge predictors overlap above 98.8%.

D. Beam Prediction on Scenario 23

Blind prediction (Table VII). At t + 1 (≈140 ms) the augmented GRU ensemble selects a beam within the 0.25 dB noise margin 96.4% of the time (class-balanced 91.7%), has 96.1% top-3 accuracy, and delivers 99.1% of the optimal received power. Its exact top-1 is 72.1% against the 79.0% oracle ceiling. At t + 3 it reaches 86.0% within noise and 96.8% power ratio. Trained on hard labels, a single GRU falls below persistence at t + 3; the soft target makes it competitive, and augmentation with ensembling makes it the best model on every metric at both horizons, including blind macro-F1 at t + 3 (0.28 vs 0.26).

TABLE VII

BLIND BEAM PREDICTION (NO BEAM MEASURED AT THE PREDICTED TIME), GROUPKFOLD OVER 35 SEQUENCES. EXACT TOP-1 AND MACRO-F1 ARE BOUNDED BY MEASUREMENT NOISE: THE LAST ROW IS A NON-CAUSAL ORACLE THAT SEES THE SWEEPS ON BOTH SIDES OF THE PREDICTED ONE. IN PARENTHESES: EXACT TOP-1 AS % OF THAT BOUND. DECISIVE / NEAR-TIE: EXACT TOP-1 ON SAMPLES WHOSE BEST BEAM LEADS THE RUNNER-UP BY MORE THAN THE 0.25 DB NOISE FLOOR (25% OF SAMPLES) AND ON THE OTHER 75%. STACKED: NESTED COMBINATION OF ALL ROWS ABOVE PLUS A SHIFTED-TEMPLATE TRACKER AND TREE MODELS; "ALL SENSING" ADDS THE GPS AND CAMERA (UAV PIXEL) RUNS OF TABLE XI. THE DEPLOYED TASK, WHICH MEASURES A FEW BEAMS, IS IN TABLES VIII–X AND XIV (BEST SCHEMES: EXACT TOP-1 ABOUT 99%)

Horizon	Model	Within noise	Balanced within noise	Top-3	Power ratio	Exact top-1	Macro-F1	Decisive	Near-tie
t+1	Persistence	93.7	86.3	94.5	98.6	69.2 (88%)	0.45	89.8	62.3
t+1	RF + sweep prior	93.7	86.7	94.9	98.6	70.4 (89%)	0.41	–	–
t+1	GRU, hard label	93.7	82.3	94.6	98.6	70.9 (90%)	0.41	93.1	63.5
t+1	GRU, soft target	95.6	89.3	95.1	98.9	70.9 (90%)	0.45	94.7	63.0
t+1	GRU ens., augmented	96.4	**91.7**	96.1	99.1	72.1 (91%)	0.48	94.1	64.7
t+1	GRU ens. + sweep, fused	96.3	90.9	96.3	99.1	72.4 (92%)	0.49	94.0	65.3
t+1	Stacked, power only	97.0	91.6	96.9	**99.2**	74.6 (94%)	0.49	95.6	67.6
t+1	**Stacked, all sensing**	**97.2**	91.1	**97.3**	**99.2**	**75.6 (96%)**	**0.50**	**96.0**	**68.9**
t+3	Persistence	78.1	66.1	80.3	93.7	53.3 (67%)	0.26	69.2	47.9
t+3	GRU, hard label	73.2	57.9	80.6	93.1	49.5 (63%)	0.21	73.6	41.5
t+3	GRU, soft target	81.1	65.8	84.8	95.5	55.3 (70%)	0.24	78.6	47.5
t+3	GRU ens., augmented	86.0	75.5	86.9	96.8	56.7 (72%)	0.28	78.7	49.4
t+3	GRU ens. + sweep, fused	85.2	74.7	86.5	96.5	57.3 (73%)	0.29	76.6	50.9
t+3	Stacked, power only	87.1	75.6	90.0	96.9	60.9 (77%)	0.32	82.8	53.6
t+3	**Stacked, all sensing**	**89.6**	**77.3**	**91.9**	**97.4**	**64.3 (81%)**	**0.34**	**86.3**	**56.9**
any	*Noise bound (non-causal oracle)*	–	–	–	–	*79.0 (100%)*	*0.59*	–	–

Where blind exact prediction fails (Table VII, `blind_prediction_analysis.py`). Exact top-1 depends almost entirely on whether the label is decided by the channel or by noise. On the 25% of samples whose best beam leads by more than the 0.25 dB noise floor, the GRU ensemble predicts the exact beam 94.1% of the time at t + 1 (95% flight-bootstrap CI 91.7–95.9%) and 78.7% at t + 3 (71.4–84.9%). On the other 75%, two or more beams are equal within the noise, the label is a coin toss among them, and every single model, including persistence, is near 62–65% at t + 1. Blind exact top-1 therefore cannot approach 95% on this data, while the measured schemes below can. Fusing the GRU ensemble with the last sweep, with the weight chosen on the other folds' flights, adds 0.3 points of exact top-1 at t + 1 (72.4%) and 0.6 at t + 3 (57.3%), and macro-F1 by 0.004 and 0.012, at the cost of 0.1–0.8 points of within-noise accuracy: the remaining gap to the 79% bound is the information in future sweeps that no causal predictor has.

Stacked blind predictor (Table VII, Fig. 5, `blind_stacking.py`). No single model closes that gap, but the models make different errors, so we combine them. The components are the GRU variants of every side-information run (power only, GPS, UAV pixel, both), extra trees and LightGBM on long-history features (eight sub-beam peak positions and their steps, four sweeps, the lobe around the peak, side information), the last sweep, and a shifted template: the UAV's sub-beam peak (parabolic fit around the argmax) is tracked over the recent sweeps, the last sweeps are aligned on it and averaged, and the average is shifted by the predicted peak motion. Each component's scores are standardised per sample and summed with weights found by coordinate ascent on the four training folds (within-noise accuracy plus half the exact top-1), together with the template's three settings, and applied unchanged to the held-out fold. With power sweeps alone the stack raises exact top-1 from 72.1 to 74.6% at t + 1 and from 56.7 to 60.9% at t + 3. With all sensing it reaches 75.6% [72.4, 78.4] at t + 1, 96% of the noise bound, and 64.3% [59.5, 68.8] at t + 3, with 97.2% and 89.6% within noise. On decisive labels it predicts the exact beam 96.0% of the time at t + 1 and 86.3% at t + 3. Over the 30 flights long enough for an eight-sweep window, the all-sensing stack beats the GRU ensemble by 3.6 points at t + 1 [1.9, 5.1] (24 flights better, 4 worse) and 7.6 points at t + 3 [5.6, 9.9] (23 better, 6 worse); the power-only stack by 2.5 [0.9, 3.8] and 4.2 [2.7, 6.2] points (Wilcoxon, Holm $p \leq 0.003$ for all four; Table XIII). The largest weights go to the augmented ensembles of the power-only, GPS and GPS + UAV pixel runs, and sensing adds significantly inside the stack: the all-sensing stack beats the power-only stack by 1.1 points at t + 1 [0.6, 1.7] (21 better, 4 worse; Holm $p = 0.006$) and 3.4 at t + 3 [1.7, 5.1] (19 better, 8 worse; $p = 0.045$). Position information that is not significant in any single model (Table XIII) therefore becomes useful once it is combined with the others. Adding the two camera-embedding runs of Table XI to the stack changes exact top-1 by at most 0.1 points, and they receive small weights, so the stack uses the four runs above. Exact top-1 is still below 95%: in 75% of sweeps the label is decided by noise (above).

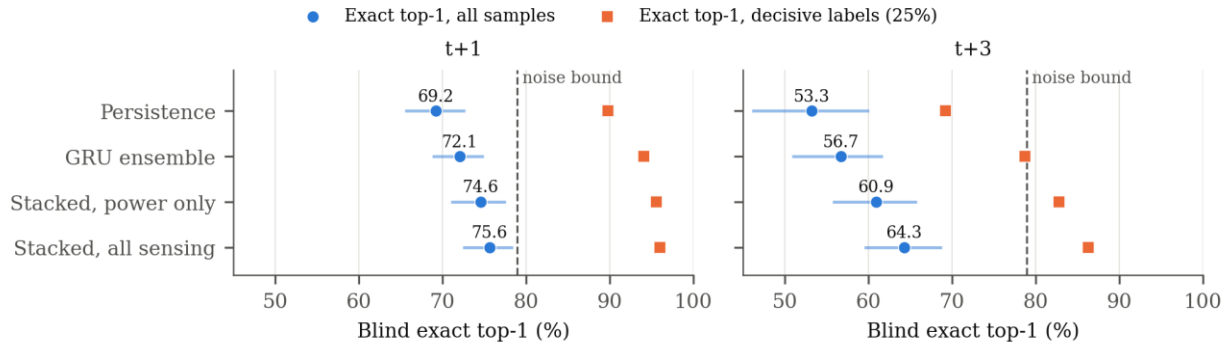


Fig. 5. Blind exact top-1 per model on all samples (circles, 95% flight-bootstrap CI) and on decisive labels (squares), against the non-causal noise bound (dashed). Stacked: nested combination of Table VII.

Reduced beam training (Table VIII, Fig. 6). Measuring a handful of predicted beams resolves the near-ties that blind prediction cannot. At t + 1, the GRU ensemble with the tie graph selects the exact optimal beam 99.3% of the time with macro-F1 0.961 over all 56 classes, measuring 7 of 64 beams in 0.55 ms; the 95% interval of exact top-1 lies above 95%. At t + 3, the same scheme reaches 99.1% exact top-1 and macro-F1 0.974 over beams with at least five samples, with 11.7 beams. The single-burst variant reaches 98.9% / 0.971 in 1.22 ms. Every scheme in Table VIII also keeps at least 99.5% of decisions within the 0.25 dB noise floor and at least 99.8% of the best beam's power. The stacked blind predictor of Table VII also shortens the shortlist: at t + 1 the power-only stack, in a single round, matches the two-round GRU scheme (99.3%, macro-F1 0.961) with 6.4 instead of 7.0 beams and no second round (0.50 ms). At t + 3 the all-sensing stack reaches 99.3% exact top-1 with 14.8 beams in one round (1.16 ms) and, with the tie graph, 99.4% and all-class macro-F1 0.935; the rarely seen beams keep the latter below 0.95.

TABLE VIII

REDUCED BEAM TRAINING (OPEN-LOOP: FULL-SWEEP HISTORY ASSUMED), NESTED-CV OPERATING POINTS, SCORED AGAINST THE FULL-SWEEP GROUND TRUTH [95% CI OVER SEQUENCES]. LATENCY: BEAM-TRAINING TIME UNDER 5G NR FR2 TIMING (FULL SWEEP: 5 MS). MACRO-F1 ≥ 5: OVER BEAM CLASSES WITH AT LEAST FIVE SAMPLES. STACKED: THE NESTED STACK OF TABLE VII ("ALL SENSING" ADDS GPS AND THE CAMERA'S UAV PIXEL)

Horizon	Shortlist	Beams	Latency (ms)	Exact top-1 (%)	Macro-F1 (56 classes)	Macro-F1 ≥ 5
t+1	Last sweep, K = 4 (85% target)	4.2	0.33	97.6 [96.4, 98.5]	0.871 [0.829, 0.912]	0.945
t+1	Last sweep, K = 6	6.0	0.47	99.1 [98.6, 99.5]	0.960 [0.940, 0.985]	0.981
t+1	GRU ensemble + tie graph, two-round	7.0	0.55	99.3 [98.7, 99.7]	0.961 [0.941, 0.990]	0.985
t+1	GRU ensemble + last sweep, burst	6.4	0.50	99.2 [98.7, 99.6]	0.955 [0.940, 0.977]	0.984
t+1	**Stacked (power only), single round**	**6.4**	**0.50**	**99.3 [98.9, 99.7]**	**0.961 [0.943, 0.989]**	**0.986**
t+3	Last sweep, K = 16	16.0	1.25	97.9 [96.6, 98.8]	0.860 [0.823, 0.906]	0.956
t+3	GRU ensemble, single round	15.2	1.19	98.9 [98.0, 99.5]	0.896 [0.864, 0.944]	0.965
t+3	Last sweep + tie graph, two-round	13.9	2.11	99.4 [98.9, 99.7]	0.914 [0.889, 0.956]	0.979
t+3	**GRU ensemble + tie graph, two-round**	**11.7**	**1.93**	**99.1 [98.3, 99.7]**	**0.929 [0.905, 0.964]**	**0.974**
t+3	GRU ensemble + tie graph, burst	15.6	1.22	98.9 [98.1, 99.6]	0.912 [0.880, 0.949]	0.971
t+3	Stacked (all sensing), single round	14.8	1.16	99.3 [98.9, 99.7]	0.897 [0.867, 0.965]	0.978
t+3	Stacked (all sensing) + tie graph, two-round	13.6	2.09	99.4 [98.9, 99.8]	0.935 [0.910, 0.972]	0.979

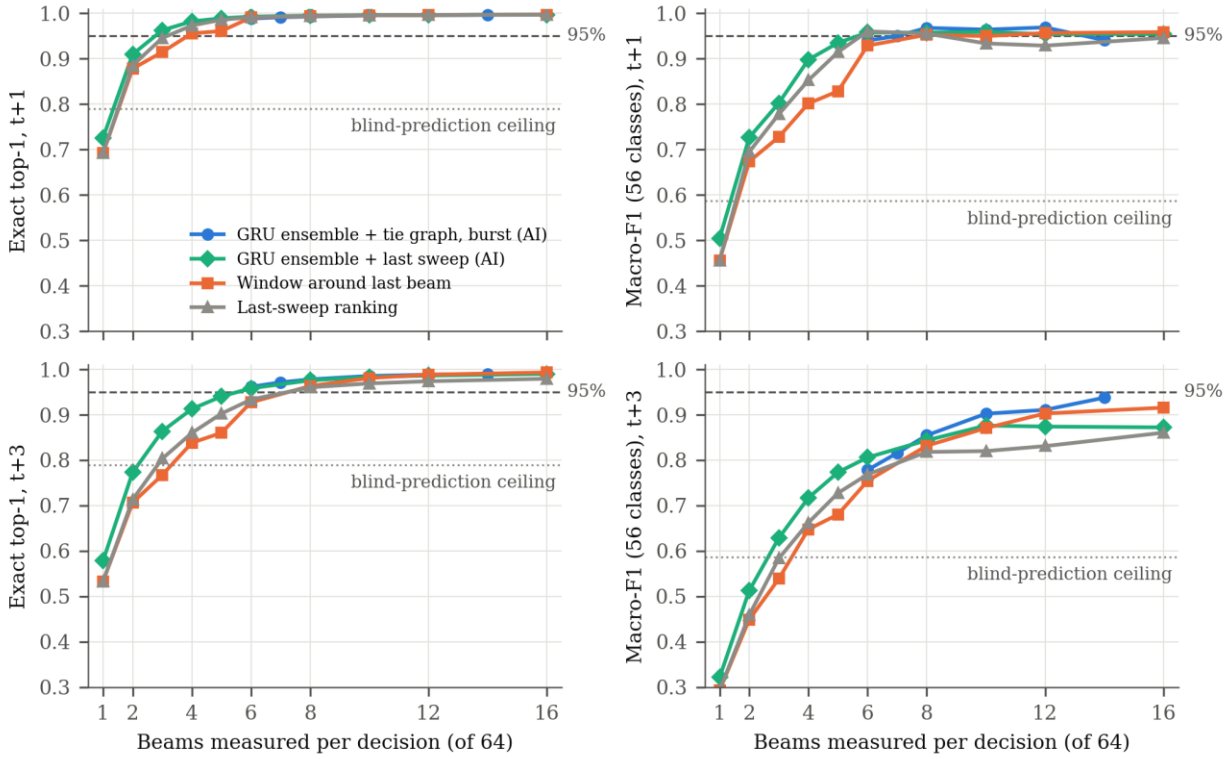


Fig. 6. Exact top-1 (left) and all-class macro-F1 (right) versus beams measured at t + 1 (top) and t + 3 (bottom), out-of-fold predictions; dashed: 95% target; dotted: non-causal blind-prediction ceiling.

Raising the target from 85% to 95% costs 2–3 more beams at t + 1, still under 0.6 ms. At this horizon a non-learned 6-beam shortlist also meets the target, so the learned ranking adds little. The tie graph and the GRU matter at t + 3: all-class macro-F1 rises from 0.860 (last sweep, 16 beams) to 0.929 with 11.7 beams, and the GRU ensemble needs 16% fewer beams than the last-sweep ranking with the same graph. At t + 3, all-class macro-F1 stays below 0.95, and the reason is identifiable: three beams (23, 34 and 55) occur once in the whole dataset, and in each case the best beam beats the runner-up by about 0.01 dB. Each such class costs 1/56 ≈ 0.018 of macro-F1. Over classes with at least five samples, every t + 3 scheme with the tie graph exceeds 0.97. Pooled curves reach 0.95 all-class with 16–18 beams, but settings selected on four folds do not hold it on the fifth, so we report the nested values.

Conformal shortlists (Fig. 7). Calibrated for 95% exact top-1, the rank-based set reaches 96.1% with 3 beams at t + 1 (worst held-out fold 95.3%) and 95.5% with 5.4 beams at t + 3 (worst fold 92.7%); for a 99% target, 99.2% with 6.2 beams and 99.0% with 17.4 beams. Coverage tracks the target on average, but a single flight group can fall short by up to 2.3 points, which is the cost of non-exchangeable flights. Adaptive sets over-cover (98.8% for a 95% target at t + 1) with a 95th-percentile size of 41–56 beams, which is worse for worst-case latency.

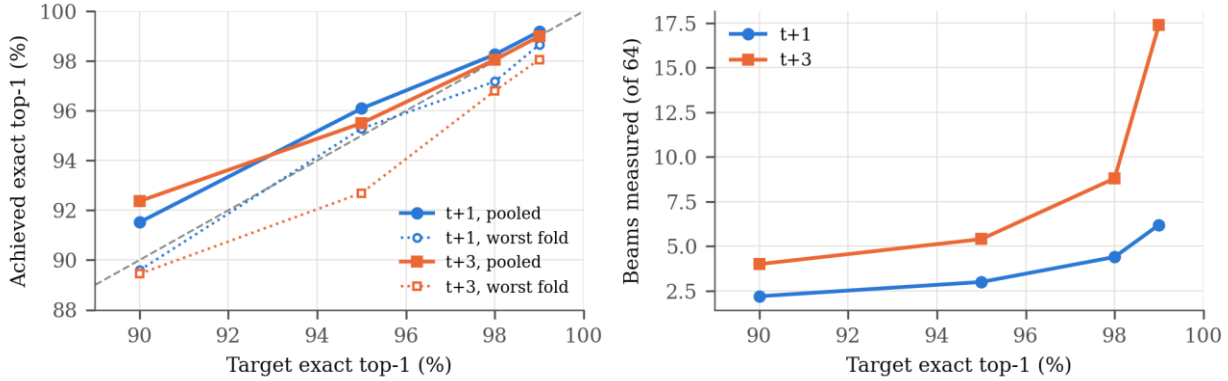


Fig. 7. Conformal shortlists (rank score, GRU ensemble). Left: achieved exact top-1 versus the target, pooled over held-out folds (solid) and for the worst fold (dotted); dashed: achieved = target. Right: mean beams measured.

Closed loop (Table IX). Without full-sweep history, a ±1 window with loss-triggered sweeps finds the exact beam only 75.5% of the time; adding six tie-graph partners raises this to 96.1% at 0.64 ms per decision. With a full sweep every 8 steps, tie-graph tracking reaches 97.2% exact top-1 and 99.2% within noise at 1.05 ms. At the testbed's 142 ms step a full sweep takes only 3.5% of the airtime, so effective spectral efficiency rises modestly, from 96.5% of the genie to 99.0–99.2%; the main gain is the 5× lower beam-training latency per decision.

TABLE IX

CLOSED-LOOP REPLAY OF THE 33 FLIGHTS WITH MORE THAN ONE SWEEP (FLIGHTS 22 AND 30 HOLD A SINGLE SWEEP); EACH POLICY KNOWS ONLY WHAT IT MEASURED. LATENCY: MEAN BEAM-TRAINING TIME PER DECISION. SE: EFFECTIVE SPECTRAL EFFICIENCY AS % OF A GENIE THAT ALWAYS USES THE BEST BEAM WITH NO TRAINING TIME, AT NOMINAL BEST-BEAM SNR OF 0 / 10 / 20 DB

Policy	Exact top-1 (%)	Within noise (%)	Latency (ms)	SE (% of genie)
Full sweep every step	100	100	5.00	96.5 / 96.5 / 96.5
Window ±1, full sweep every 4	91.2	96.4	1.50	98.2 / 98.5 / 98.7
Window ±1 + 4 tie partners, full every 4	**97.8**	**99.4**	1.61	98.7 / 98.8 / 98.8
Window ±1 + 4 tie partners, full every 8	**97.2**	**99.2**	**1.05**	**99.0 / 99.1 / 99.2**
Window ±1, loss-triggered sweeps only	75.5	83.2	0.35	92.9 / 95.0 / 96.8
Window ±1 + 6 tie partners, loss-triggered only	96.1	96.9	0.64	98.5 / 98.9 / 99.2

Learned tracking in the loop (Table X, Fig. 8). At equal budget, every learned model is below the tie graph. With two extra beams, the tracker with six tie partners reaches 99.16% exact top-1 at 1.18 ms, while the ten models reach 98.17–98.71%; with four extra beams, 99.28% against 98.59–98.83%. The models differ from each other by at most 0.6 points, so architecture and the training settings we varied matter little. The configuration selected on validation flights (the Transformer, 91.9% validation top-4) is not better on test than the others. The extra beams of every model still help over the four-partner tracker (97.15%), so the models learn useful structure, but the tie graph already captures it. For latency, the MLP is fastest (0.11 ms fp32); dynamic int8 quantization made every model slower on this CPU and is not supported for the Transformer encoder in our PyTorch version. Inference runs between steps, off the beam-training critical path.

TABLE X

LEARNED CLOSED-LOOP TRACKING (HELD-OUT FLIGHTS, POOLED OVER FOLDS). EXACT TOP-1 WITH THE TRACKER (FOUR PARTNERS) PLUS K MODEL BEAMS; TIE-GRAPH BASELINE AT THE SAME BUDGET: 99.16% (K = 2, 1.18 MS) AND 99.28% (K = 4, 1.31 MS). INFERENCE: MEDIAN BATCH-1 LATENCY, ONE LAPTOP CPU THREAD

Model	Val. top-4 (%)	Exact top-1, k = 2 (%)	Exact top-1, k = 4 (%)	fp32 (ms)	int8 (ms)
Tie graph, +k partners (no learning)	–	**99.16**	**99.28**	–	–
GRU, h = 128	90.3	98.37	98.71	0.54	0.88
LSTM, h = 128	89.8	98.51	98.78	0.62	1.24
Temporal CNN, h = 128	89.8	98.47	98.75	0.67	0.80
Transformer, h = 64 (selected)	91.9	98.39	98.71	1.00	n/a
MLP, h = 256	90.5	98.48	98.81	**0.11**	0.33
GRU, h = 64	89.4	98.24	98.68	0.43	0.87
GRU, lr = $3 \cdot 10^{-4}$	90.0	98.24	98.70	0.50	0.92
GRU, W = 4	90.3	98.17	98.59	0.33	0.64
GRU, W = 16	89.3	98.29	98.64	0.91	1.67
GRU, hard target	91.2	98.71	98.83	0.56	1.04

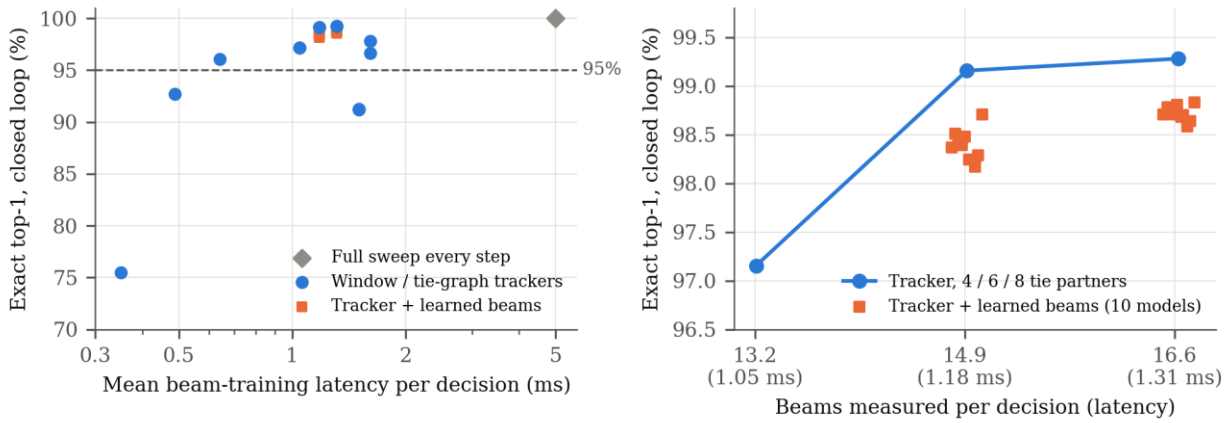


Fig. 8. Closed loop. Left: exact top-1 versus mean beam-training latency for every policy (log scale); dashed: 95%. Right: at equal budget, extra beams chosen by the tie graph (line) versus by each of the ten learned models (squares).

Position- and vision-aided prediction (Tables XI–XII, Fig. 9). The full Scenario 23 release adds, for every sample, the UAV's GPS position, speed, velocity components, height, pitch and roll. We convert the GPS fix to east/north offsets from the base station and its bearing, append these and the raw kinematic channels to every step of each model's input window, and repeat the blind-prediction study with the same folds. All other tables use power sweeps only, so that they can be reproduced from the beam CSV shipped with the repository. The altitude and height channels are in feet, not metres: altitude minus height is the same 1175.8 ft for every sample, the ground elevation at the base station (West Chandler, AZ), so we convert both to metres. The UAV then flies 9.6–119 m above its take-off point, consistent with the 120 m range of the dataset, and its elevation seen from the base station is 46–90° (median 77°).

The base station also records one 960 × 540 camera frame per sweep, looking up at the sky; the UAV appears as a dark blob of a few pixels, often next to a sun-glare streak. We use the frames in two ways. (i) Generic features: every frame is encoded by a ResNet-18 pretrained on ImageNet and kept frozen (input 224 × 398, 512-d global-pooled embedding), classified with a logistic regression or a one-hidden-layer MLP, or appended to every step as a 16-component PCA (fitted on all frames without labels, 79% of the variance). (ii) UAV localisation without labels: the frame minus a Gaussian-blurred copy of itself ($\sigma = 12$ px) removes the smooth sky, and the most negative 3 × 3 mean of the residual is the UAV. Its pixel (x, y) and contrast form a three-number feature. In 95% of frames the UAV is at least 29 grey levels darker than the surrounding sky; inspected frames show the detection on the UAV body, also next to the glare and when the UAV is close. We also check the locator against GPS, without labels: a quadratic map from pixel to the UAV's east/north offset, fitted on other flights, explains 67% of the east and 12% of the north variance (median error 4.5 m). The UAV is high above the camera (elevation 46–90°), so a pixel fixes the direction but not the distance, which also depends on the 9.6–119 m height; in the frames with the largest disagreement the detection is visibly on the UAV. The raw pixel jumps by more than 100 px between consecutive frames in 1.8% of frame pairs, while the GPS moves at most 2.2 m (99th percentile), so about 1% of frames are misdetected. A causal gate removes half of these jumps: a detection more than 100 px from the last accepted one is replaced by that position until two consecutive detections agree on the new place. The gate holds 1.4% of frames and leaves 0.95% of frame pairs with a jump above 100 px; all results below use the gated pixel. The locator takes 54 ms per frame on one CPU process, most of it JPEG decoding; the ResNet-18 encoder takes 108 ms on one thread.

Alone, the UAV pixel identifies the current beam within noise 80.4% of the time (58.5% exact), against 67.4% for the generic embedding and 14.9% for the most frequent beam (Table XII), so localising the UAV is what turns the image into beam information. For blind prediction from power sweeps, however, no side information gives a significant gain (Table XI, Table XIII). Position helps most further ahead: at t + 3, pooled within-noise accuracy rises by 4.2 points for the random forest (78.5 → 82.7%) and by 1.4 for the GRU ensemble (86.0 → 87.4%), and by 1.3 with GPS and the UAV pixel together (87.3%), but per flight these gains are not significant on their own (16 flights better, 13 worse; Holm $p = 0.64$). Combined with the power-only models in the stack of Table VII, the same runs do add significantly. At t + 1 every change is below one point: the last sweep already encodes where the UAV is. The generic embedding hurts: it lowers every model at both horizons, the GRU ensemble at t + 3 by 5.6 points (worse on 23 of 30 flights, Holm $p < 10^{-4}$), because with 35 flights 16 scene features let the models fit flight-specific appearance that does not transfer. Exact top-1 changes by −5.0 to +3.6 points across configurations and stays far below the 79% noise ceiling in all of them, as it must.

TABLE XI

BLIND BEAM PREDICTION WITH EACH SET OF SIDE INFORMATION (GROUPKFOLD OVER 35 SEQUENCES, FOLD MEANS). WITHIN NOISE / CLASS-BALANCED WITHIN NOISE / EXACT TOP-1, IN %; BOLD: BEST PER MODEL AND HORIZON. PERSISTENCE: 93.7 / 86.3 / 69.2 AT T + 1 AND 78.1 / 66.1 / 53.3 AT T + 3

Horizon	Side information	RF + sweep prior	GRU, soft target	GRU ensemble
t+1	None (power only)	93.7 / 86.7 / 70.4	**95.6 / 89.3 / 70.9**	96.4 / 91.7 / 72.1
t+1	GPS	93.9 / 85.5 / 71.0	94.9 / 86.9 / 70.0	**96.5 / 91.9 / 71.8**
t+1	Camera embedding	93.1 / 84.4 / 69.7	93.0 / 84.5 / 67.9	95.9 / 89.6 / 69.1
t+1	GPS + camera embedding	93.7 / 85.5 / 70.1	94.8 / 86.1 / 69.1	96.2 / 90.1 / 70.7
t+1	UAV pixel	93.7 / 85.5 / 70.4	94.2 / 87.1 / 69.3	96.4 / 90.8 / 71.8
t+1	GPS + UAV pixel	**94.0 / 85.0 / 71.1**	94.8 / 87.6 / 69.9	96.5 / 92.1 / 71.0
t+3	None (power only)	78.5 / 63.0 / 53.9	81.1 / 65.8 / 55.3	86.0 / 75.5 / 56.7
t+3	GPS	**82.7 / 65.3 / 57.5**	83.5 / 70.0 / 56.9	**87.4 / 78.4 / 58.3**
t+3	Camera embedding	75.9 / 61.0 / 51.2	74.8 / 59.0 / 50.3	80.5 / 67.3 / 52.9
t+3	GPS + camera embedding	81.4 / 63.9 / 57.0	78.5 / 64.6 / 52.2	85.6 / 76.4 / 56.7
t+3	UAV pixel	78.4 / 63.1 / 53.5	78.7 / 62.4 / 52.9	84.6 / 74.8 / 55.7
t+3	GPS + UAV pixel	81.5 / 64.9 / 56.8	**84.1 / 69.8 / 55.3**	87.3 / 78.5 / 58.7

TABLE XII

CAMERA ONLY: THE FRAME AT T, BEST BEAM AT T + H, NO POWER MEASUREMENT (GROUPKFOLD OVER 35 SEQUENCES)

Horizon	Camera feature, classifier	Within noise (%)	Balanced (%)	Top-3 (%)	Exact top-1 (%)	Macro-F1
t+0	None, most frequent beam	14.9	13.0	39.6	7.6	0.00
t+0	Embedding, logistic regression	63.3	62.7	70.9	40.2	0.23
t+0	Embedding, MLP	67.4	**66.5**	74.6	42.3	0.25
t+0	UAV pixel, random forest	**80.4**	65.1	**82.8**	**58.5**	**0.27**
t+1	Embedding, MLP	57.3	53.8	66.2	33.2	0.16
t+1	UAV pixel, random forest	**72.4**	**57.4**	**77.8**	**50.7**	**0.19**
t+3	Embedding, MLP	43.2	36.4	53.7	24.9	0.09
t+3	UAV pixel, random forest	**56.9**	**43.1**	**65.9**	**37.2**	**0.11**

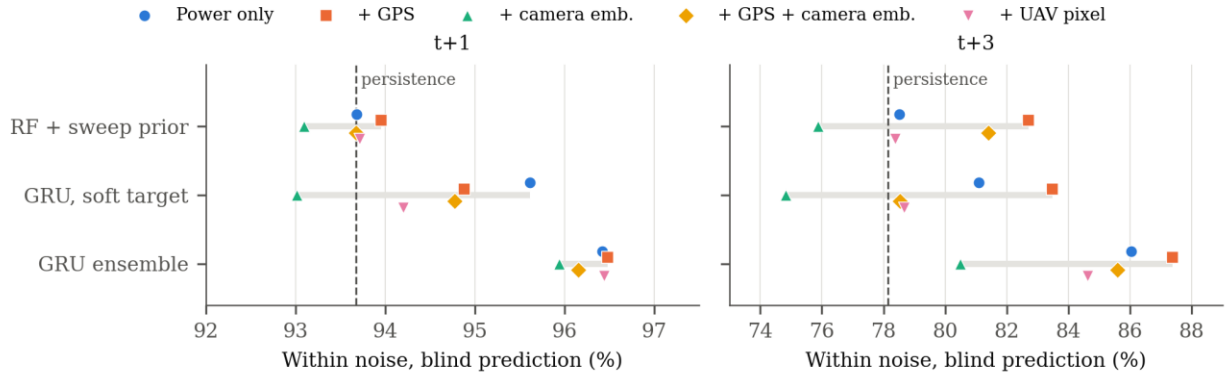


Fig. 9. Blind within-noise accuracy with each set of side information; dashed: persistence. Each series has its own marker shape.

Position-aided beam training. Rerunning the reduced beam training and conformal studies on the GPS-aided GRU ensemble changes little. At t + 3 the two-round scheme reaches 99.2% exact top-1 [98.5, 99.7] and all-class macro-F1 0.931 [0.896, 0.962] (power only: 99.1% and 0.929), but measures 15.5 instead of 11.7 beams (2.23 vs 1.93 ms); fused with the last sweep it reaches 99.4% [98.9, 99.7] and 0.955 [0.936, 0.981] with 13.8 beams (power only: 99.3% and 0.913 with 13.3). All intervals overlap. At t + 1 every operating point stays within 0.3 points. The 95% conformal shortlist at t + 3 uses 5.2 instead of 5.4 beams and its worst fold rises from 92.7 to 94.3% coverage. Position raises all-class macro-F1 at t + 3, but its lower confidence bound stays below 0.95: the singleton classes remain.

Significance (Table XIII, Fig. 10). Across the 33 replayed flights, the tie-graph tracker (four partners, full sweep every 8 steps) loses 2.8 points of exact top-1 against a full sweep every step on 27 flights and ties on 6, yet gains 2.6 points of effective spectral efficiency at 10 dB on 32 of 33 flights, because it spends 1.05 instead of 5 ms on training. Six partners improve exact top-1 by 2.0 points over four on 26 flights and lose on none; eight add nothing significant over six. Periodic sweeps every 4 instead of 8 steps buy 0.65 points of accuracy for 0.33 points of efficiency. Loss-triggered sweeps with six partners improve the median flight (22 better, 4 worse) but

lose heavily on a few, so the pooled difference is negative with a CI that includes zero; this policy is not uniformly better.

TABLE XIII

PAIRED PER-FLIGHT COMPARISONS (A − B, PERCENTAGE POINTS; 95% FLIGHT-BOOTSTRAP CI; WILCOXON SIGNED-RANK, HOLM-CORRECTED WITHIN FAMILY; FLIGHTS WHERE A IS BETTER / TIED / WORSE). BLIND ROWS: 30 FLIGHTS LONG ENOUGH FOR AN EIGHT-SWEEP WINDOW; THE BLIND FAMILY ALSO INCLUDES EXACT TOP-1 ROWS (IN `SIGNIFICANCE_TESTS.JSON`), NONE SIGNIFICANT EXCEPT THE CAMERA EMBEDDING (WORSE, HOLM $p \le 0.04$)

Family	A	B	Metric	A − B [95% CI]	Holm p	Better / tied / worse
Closed loop	4 partners, full /8	Full sweep every step	Exact top-1	−2.85 [−4.04, −1.82]	$< 10^{-4}$	0 / 6 / 27
Closed loop	4 partners, full /8	Full sweep every step	SE, 10 dB	+2.61 [+2.50, +2.71]	$< 10^{-4}$	32 / 1 / 0
Closed loop	6 partners, full /8	4 partners, full /8	Exact top-1	+2.00 [+1.33, +2.77]	$< 10^{-4}$	26 / 7 / 0
Closed loop	8 partners, full /8	6 partners, full /8	Exact top-1	+0.12 [−0.02, +0.28]	0.08	7 / 24 / 2
Closed loop	4 partners, full /4	4 partners, full /8	Exact top-1	+0.65 [+0.39, +0.96]	$< 10^{-4}$	18 / 15 / 0
Closed loop	4 partners, full /4	4 partners, full /8	SE, 10 dB	−0.33 [−0.38, −0.27]	$< 10^{-4}$	2 / 1 / 30
Closed loop	4 partners, full /8	Window ±1, full /4	Exact top-1	+5.95 [+4.28, +7.73]	$< 10^{-4}$	30 / 3 / 0
Closed loop	6 partners, loss-triggered	4 partners, full /8	Exact top-1	−1.08 [−4.73, +1.41]	0.01	22 / 7 / 4
Blind t+1	GRU ens. + GPS	GRU ens., power	Within noise	+0.05 [−0.26, +0.34]	1.0	10 / 10 / 10
Blind t+1	GRU ens. + camera emb.	GRU ens., power	Within noise	−0.49 [−0.78, −0.21]	0.020	4 / 9 / 17
Blind t+1	GRU ens. + GPS + UAV pixel	GRU ens., power	Within noise	+0.03 [−0.26, +0.30]	1.0	7 / 12 / 11
Blind t+3	GRU ens. + GPS	GRU ens., power	Within noise	+1.33 [−0.02, +2.77]	0.64	16 / 1 / 13
Blind t+3	GRU ens. + camera emb.	GRU ens., power	Within noise	−5.55 [−7.88, −3.29]	$< 10^{-4}$	2 / 5 / 23
Blind t+3	GRU ens. + GPS + camera emb.	GRU ens., power	Within noise	−0.44 [−1.94, +1.00]	0.64	10 / 5 / 15
Blind t+3	GRU ens. + UAV pixel	GRU ens., power	Within noise	−1.41 [−3.18, −0.12]	0.64	9 / 5 / 16
Blind t+3	GRU ens. + GPS + UAV pixel	GRU ens., power	Within noise	+1.24 [−0.28, +2.70]	0.64	18 / 2 / 10
Stacking t+1	Stacked, all sensing	GRU ens., power	Exact top-1	+3.59 [+1.89, +5.07]	< 0.001	24 / 2 / 4
Stacking t+1	Stacked, all sensing	Stacked, power only	Exact top-1	+1.09 [+0.56, +1.67]	0.006	21 / 5 / 4
Stacking t+3	Stacked, all sensing	GRU ens., power	Exact top-1	+7.58 [+5.61, +9.89]	0.002	23 / 1 / 6
Stacking t+3	Stacked, all sensing	Stacked, power only	Exact top-1	+3.35 [+1.73, +5.09]	0.045	19 / 3 / 8

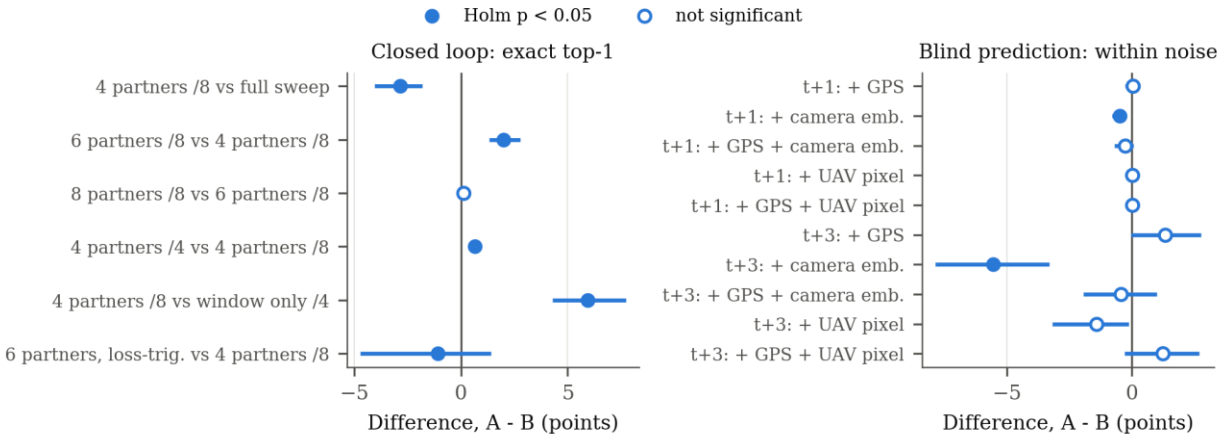


Fig. 10. Paired per-flight differences with 95% flight-bootstrap CIs. Filled: Wilcoxon signed-rank on per-flight differences, Holm $p < 0.05$. The two can disagree when most flights improve but a few lose heavily (bottom left).

Sensing-aided closed-loop tracking (Table XIV, Fig. 11). Side information that does not help blind prediction can still help the tracker, because there it only has to propose candidates that are then measured. In the closed-loop replay of Table IX we add, at every step, the top k beams of a random forest that maps the previous step's sensing (UAV pixel, GPS, or both) to the current best beam. Using the previous frame keeps the predictor causal and leaves one sweep interval for image processing. The predictor and the tie graph are fitted on the other folds' flights. The extra candidates replace the periodic full sweeps: a full sweep now follows only a beam loss.

With camera and GPS candidates (k = 4), the tracker selects the exact best beam 99.24% of the time with 7.0 beams and 0.56 ms per decision, against 97.15% with 13.2 beams and 1.05 ms for the tie-graph tracker with periodic sweeps: better on 25 flights and worse on 1, with lower latency on 30 of 33 flights. It matches the tracker with eight tie partners (99.28%) at 43% of its latency (1.31 ms), and delivers 99.59% of the genie's spectral efficiency at 10 dB, within 0.03 points of the best policy (99.62%, k = 2). The camera alone (98.44%, 0.59 ms) and GPS alone (99.12%, 0.59 ms) are each enough for most of the gain. Without side information, dropping the periodic sweeps costs 4.5 points (92.70%), so the candidates

are what make loss-triggered sweeps safe. Without tie partners, camera candidates are not enough (96.08%): the camera locates the UAV, the tie graph resolves which of the beams near that direction is strongest. Full sweeps remain for beam loss in 1.8% of steps, so the 95th-percentile latency of the worst flight is 6 ms. The gain extends to rare beams: macro-F1 rises from 0.934 to 0.984 over beams with at least five steps and from 0.843 to 0.951 over all 56 classes. Rare beams are what limit all-class macro-F1, and a few more candidates reach them: six camera + GPS candidates give 99.46% exact top-1 and all-class macro-F1 0.972 with 8.2 beams (0.66 ms), and four candidates with six tie partners give 99.57% and 0.963 / 0.992 with 8.4 beams (0.67 ms), better than the tracker on 27 flights and worse on none. In closed loop the sensing-aided tracker therefore meets the 95% target on exact top-1, within-noise accuracy and macro-F1 over all classes, at 13% of the latency of a full sweep.

Stale GPS and the latency model. The camera is at the base station, but the GPS fix has to be sent by the UAV. When it arrives 2, 4 or 8 steps late (0.28, 0.57 and 1.1 s), GPS top-4 candidates give 99.16, 98.69 and 98.10% exact top-1; with the camera, which stays one frame old, 99.17, 98.97 and 98.84%. Every one of these stays above the tracker with periodic sweeps (Holm $p < 0.001$), so the scheme degrades gracefully with GPS delay and the camera carries it when GPS is late. The 5G NR FR2 timing is an assumption too, so we rescaled each replayed decision with a per-beam time of 39, 78 or 156 µs and a report round trip of 0.5, 1 or 2 ms. Camera + GPS top-4 then needs 0.28–1.13 ms per decision, against 0.52–2.09 ms for the tracker and 2.5–10 ms for full sweeps, and keeps the highest effective spectral efficiency of the three in all nine cases (99.2–99.8% of the genie). The ordering of the policies does not depend on the timing values.

TABLE XIV

CLOSED-LOOP REPLAY OF 33 FLIGHTS WITH CAMERA (UAV PIXEL) AND GPS CANDIDATES; "+ T": WITH THE FOUR-PARTNER TRACKER ("T6": SIX PARTNERS); "LOSS": FULL SWEEPS ONLY ON BEAM LOSS (1.8% OF STEPS; TRACKER WITH FULL EVERY 8: 13.9%). MACRO-F1 OVER ALL BEST-BEAM CLASSES / CLASSES WITH AT LEAST FIVE STEPS. MS: MEAN BEAM-TRAINING TIME PER DECISION (95TH PERCENTILE ON THE WORST FLIGHT: 6.0 MS, FULL SWEEP EVERY STEP 5.0 MS). SE: EFFECTIVE SPECTRAL EFFICIENCY AT 10 DB, % OF GENIE. Δ: CHANGE IN EXACT TOP-1 AGAINST THE TRACKER WITH FULL SWEEPS EVERY 8 STEPS (POINTS, 95% FLIGHT-BOOTSTRAP CI); B / T / W: FLIGHTS BETTER / TIED / WORSE; HOLM $p \le 0.01$ EXCEPT (A), $p = 0.06$

Policy	Beams	ms	Top-1 (%)	Macro-F1 (all / ≥ 5)	In noise (%)	SE (%)	Δ top-1 [95% CI]	B / T / W
Full sweep every step	64.0	5.00	100.00	1.000 / 1.000	100.00	96.51	+2.85 [+1.83, +4.08]	27 / 6 / 0
Tracker, full every 8	13.2	1.05	97.15	0.843 / 0.934	99.19	99.12	reference	–
Tracker, loss	6.1	0.49	92.70	0.708 / 0.845	95.31	98.25	−4.45 [−9.35, −1.00]	2 / 18 / 13
Camera top-2 + T, loss	6.5	0.52	97.80	0.877 / 0.947	99.48	99.54	+0.65 [+0.35, +1.04]	17 / 13 / 3
Camera top-4 + T, loss	7.4	0.59	98.44	0.893 / 0.959	99.64	99.52	+1.28 [+0.82, +1.86]	24 / 9 / 0
Camera top-4, window only, loss (a)	6.2	0.50	96.08	0.776 / 0.877	98.95	99.52	−1.07 [−1.90, −0.30]	8 / 4 / 21
GPS top-2 + T, loss	6.5	0.52	98.62	0.930 / 0.967	99.78	99.61	+1.47 [+0.91, +2.14]	24 / 8 / 1
GPS top-4 + T, loss	7.3	0.59	99.12	0.944 / 0.979	**99.89**	99.58	+1.97 [+1.26, +2.76]	25 / 7 / 1
Camera + GPS top-2 + T, loss	6.3	**0.51**	98.71	0.932 / 0.972	99.81	**99.62**	+1.55 [+0.96, +2.27]	24 / 8 / 1
Camera + GPS top-4 + T, loss	7.0	0.56	99.24	0.951 / 0.984	99.88	99.59	+2.08 [+1.31, +2.97]	25 / 7 / 1
Camera + GPS top-6 + T, loss	8.2	0.66	99.46	**0.972** / 0.988	**99.89**	99.53	+2.31 [+1.48, +3.29]	26 / 6 / 1
Camera + GPS top-4 + T6, loss	8.4	0.67	**99.57**	0.963 / **0.992**	**99.89**	99.52	**+2.42 [+1.59, +3.36]**	**27 / 6 / 0**

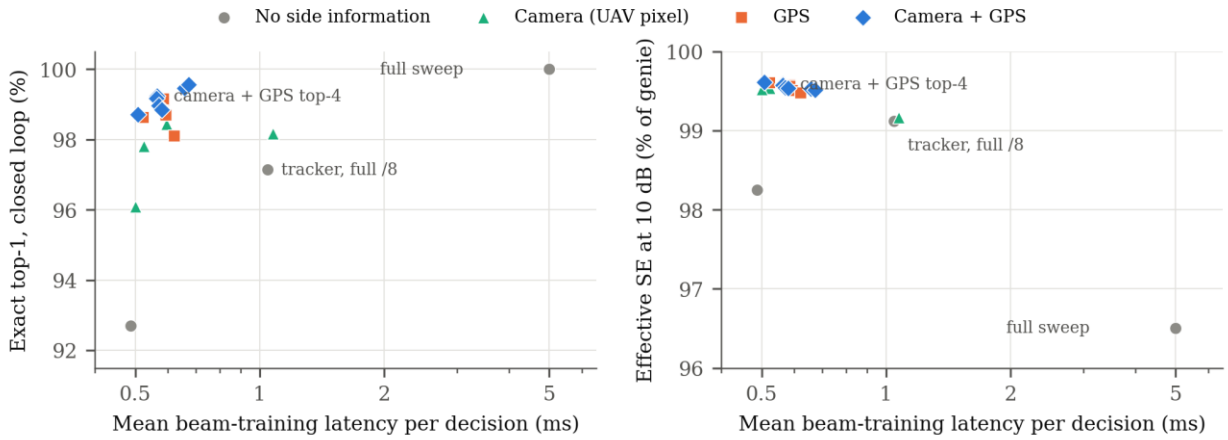


Fig. 11. Closed loop with camera and GPS candidates: exact top-1 (left) and effective spectral efficiency at 10 dB (right) against mean beam-training latency (log scale).

VII. Discussion

Metrics shape conclusions. Two conclusions of our earlier draft reversed under correct metrics. A Random Forest reported at 80.9% on the current-beam task was below the 100% that argmax achieves, and "open-loop phase tracking is impossible at sub-THz" followed from per-element phase error, which is blind to the common phase offset. We recommend that RIS tracking studies report delivered gain and that beam-prediction studies on argmax-labelled data predict future beams under sequence-level splits.

Where hardware awareness pays. The largest impairment is amplitude–phase coupling. Coupling-aware control recovers 0.55 dB at 2 bits and about 1 dB at 3–4 bits; a coupled element cannot reflect strongly at every phase, so a gap remains. The estimation side gains more (LMMSE removes the coupling bias), and the largest system-level effects are the RIS sizes needed for coverage: 2.25× the ideal element count at 60 GHz, and a 24–32 cm panel at 140 GHz once D-band transceiver limits are included, depending mainly on the receiver noise figure.

Learning where it helps. On power sweeps alone the learned gain over persistence is modest at t + 1 and clear at t + 3 (86.0% vs 78.1% within noise). In reduced beam training, the largest single gain comes from a simple learned structure, the beam-tie graph, rather than from the sequence model. The GRU then reduces the beams measured at t + 3. In closed loop, where only measured beams are known, ten deep sequence models trained on those measurements did not beat the tie graph at equal budget (Table X). On this dataset, the useful structure is which beams tie, and a co-occurrence count captures it; we report this negative result because it bounds what learning adds once measurement is cheap. The soft target and beam-axis augmentation matter more than the architecture. Logit adjustment for the class imbalance [19] lowered every metric because the beam distribution differs between flights. Side information follows the same pattern for blind prediction: in any single model UAV position gives a small, not significant gain at 0.4 s, and generic camera features hurt, because 35 flights are too few to learn which parts of a scene matter. Combining models helps more than any one of them: the nested stack gains 3.6 points of exact top-1 at t + 1 and 7.6 at t + 3, and inside it position and the located UAV add a significant 1.1 and 3.4 points. The same sensing pays off as soon as its output is measured rather than trusted: as candidates in the tracker, the located UAV and its GPS halve the training latency and raise exact top-1 on 25 of 33 flights, with one worse, and two more candidates bring the rarely seen beams above 0.95 macro-F1 (Table XIV). Making the camera useful took a hand-built locator, not a larger model.

Limitations, and what was done about each. (i) *Hardware coefficients.* The Saleh and coupling coefficients are literature values, not measurements of a 60 GHz PA or RIS. Table II bounds the effect of the coupling strength, and Sec. VI-A shows that the conclusions hold for PA back-off of 3–12 dB, obliquity exponents of 0–2 and D-band transmit power of 11–17 dBm. Measured coefficients need hardware we do not have. (ii) *Simulation.* The physical-layer results are simulated; only the beam-management results use measurements. (iii) *Inputs.* Tables VII and XI–XIV use GPS and camera; the other tables use power sweeps only, so that they can be reproduced from the beam CSV shipped with the repository. (iv) *Locator.* It assumes a sky background and is checked against GPS (median 4.5 m) and by inspection, not against labelled positions, which Scenario 23 does not provide; after the causal gate, 0.95% of frame pairs still jump by more than 100 px. (v) *GPS delay and timing.* Both were assumptions of the first replay; the stale-GPS and timing sensitivity in Sec. VI-D shows that neither changes the conclusions. (vi) *Height unit.* The dataset does not document it; the constant altitude–height offset shows that it is feet, and we convert it. (vii) *Rare beams.* Seven of the 56 best-beam classes have fewer than five samples (10 samples, 0.09% of the data) but carry 12.5% of the macro-F1 weight, and five occur in a single flight. They limit all-class macro-F1 of blind prediction and of reduced beam training at t + 3 (0.93); in closed loop, two more candidates reach them (0.97). Estimating their performance reliably needs more flights, not a different model. (viii) *Blind exact top-1* is bounded at 79% by measurement noise; no method on this data can reach 95% without measuring beams. (ix) *Open loop.* Table VIII assumes full-sweep history; Tables IX, X and XIV give the closed-loop figures. (x) *SNR.* Effective spectral efficiency is reported at nominal SNRs because the dataset's powers are unitless.

VIII. Conclusion

We built a hardware-aware model of an RIS-assisted mmWave UAV link and used it, with real 60 GHz measurements, to evaluate beam management end to end. Hardware impairments cost 7.8 dB and 2.25× RIS elements at 60 GHz, and D-band transceiver limits push the 140 GHz panel to 24–32 cm, depending on the receiver noise figure; a position-parametric estimator matches perfect channel knowledge with 64× fewer pilots; Kalman-based prediction preserves 99.3% of the array gain at 500 ms latency; and AI-assisted reduced beam training with a learned beam-tie graph selects the exact optimal beam 99.3% of the time given full-sweep history and 97.2–99.3% in a closed-loop replay of 33 flights, with 1.05–1.31 ms of beam training per decision instead of 5 ms; with candidates from the UAV's GPS and its position in the camera image it reaches 99.2–99.6% with 0.56–0.67 ms and no periodic sweeps, with macro-F1 0.97 over all 56 beam classes for six candidates, and stays above 98.8% when the GPS is 1.1 s old. Deep sequence models trained on measured-only data did not improve on the tie graph at equal budget. For blind prediction, a nested stack of all models reaches 75.6% exact top-1 at 140 ms, 96% of the noise bound; position and camera features add significantly only inside this stack, and frozen generic camera features lower every single model. All reported metrics are evaluated under sequence-level cross-validation. Future work will validate the camera locator against labelled UAV positions, add flights to estimate rare-beam performance, and use measured hardware coefficients.

REFERENCES

[1] S. M. Shah *et al.*, "TeraHertz communication and associated challenges in 6G cellular networks," in *Proc. ICCIS*, 2021, doi: 10.1109/ICCIS54243.2021.9676382.

[2] C. Chaccour, M. N. Soorki, W. Saad, M. Bennis, P. Popovski, and M. Debbah, "Seven defining features of terahertz (THz) wireless systems: A fellowship of communication and sensing," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 967–993, 2022, doi: 10.1109/COMST.2022.3143454.

[3] G. Charan *et al.*, "Towards real-world 6G drone communication: Position and camera aided beam prediction," in *Proc. IEEE GLOBECOM*, 2022, pp. 2951–2956.

[4] V. A. Nugroho and B. M. Lee, "GPS-aided deep learning for beam prediction and tracking in UAV mmWave communication," *IEEE Access*, vol. 13, pp. 117065–117077, 2025.

[5] J. Dai, F. Zhu, C. Pan, H. Ren, and K. Wang, "Statistical CSI-based transmission design for reconfigurable intelligent surface-aided massive MIMO systems with hardware impairments," *IEEE Wireless Commun. Lett.*, vol. 11, no. 1, pp. 38–42, Jan. 2022.

[6] M. Meles, A. Rajasekaran, L. Mela, and R. Jäntti, "RIS-assisted three-dimensional drone localization and tracking under hardware impairments," *IEEE Access*, vol. 12, pp. 81348–81361, 2024.

[7] M. Banagar and H. S. Dhillon, "Fundamentals of wobbling and hardware impairments-aware air-to-ground channel model," *IEEE Trans. Veh. Technol.*, vol. 73, no. 12, pp. 17946–17962, Dec. 2024.

[8] Recommendation ITU-R P.676-13, "Attenuation by atmospheric gases and related effects," ITU, 2022.

[9] A. A. M. Saleh, "Frequency-independent and frequency-dependent nonlinear models of TWT amplifiers," *IEEE Trans. Commun.*, vol. COM-29, no. 11, pp. 1715–1720, Nov. 1981.

[10] A. Alkhateeb *et al.*, "DeepSense 6G: A large-scale real-world multi-modal sensing and communication dataset," *IEEE Commun. Mag.*, vol. 61, no. 9, pp. 122–128, Sep. 2023.

[11] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[12] K. Cho *et al.*, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. EMNLP*, 2014.

[13] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, 1960.

[14] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Commun. Mag.*, vol. 58, no. 1, pp. 106–112, Jan. 2020.

[15] S. Abeywickrama, R. Zhang, Q. Wu, and C. Yuen, "Intelligent reflecting surface: Practical phase shift model and beamforming optimization," *IEEE Trans. Commun.*, vol. 68, no. 9, pp. 5849–5863, Sep. 2020.

[16] W. Tang *et al.*, "Wireless communications with reconfigurable intelligent surface: Path loss modeling and experimental measurement," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 421–439, Jan. 2021.

[17] J. J. Bussgang, "Crosscorrelation functions of amplitude-distorted Gaussian signals," MIT Research Laboratory of Electronics, Tech. Rep. 216, 1952.

[18] E. Björnson, J. Hoydis, and L. Sanguinetti, "Massive MIMO networks: Spectral, energy, and hardware efficiency," *Found. Trends Signal Process.*, vol. 11, no. 3–4, pp. 154–655, 2017.

[19] A. K. Menon *et al.*, "Long-tail learning via logit adjustment," in *Proc. ICLR*, 2021.

[20] A. S. H. Ahmed, M. Seo, A. A. Farid, M. Urteaga, J. F. Buckwalter, and M. J. W. Rodwell, "A 140GHz power amplifier with 20.5dBm output power and 20.8% PAE in 250-nm InP HBT technology," in *Proc. IEEE/MTT-S Int. Microw. Symp. (IMS)*, Los Angeles, CA, USA, 2020, pp. 492–495.

[21] E. Turkmen, A. Burak, A. Guner, I. Kalyoncu, M. Kaynak, and Y. Gurbuz, "A SiGe HBT D-band LNA with Butterworth response and noise reduction technique," *IEEE Microw. Wireless Compon. Lett.*, vol. 28, no. 6, pp. 524–526, 2018, doi: 10.1109/LMWC.2018.2831450.

[22] 3GPP TS 38.213, "NR; Physical layer procedures for control," version 17.9.0, Release 17 (ETSI TS 138 213 V17.9.0, 2024-04), clause 4.1 (candidate SS/PBCH blocks, case D: 64 blocks per 5 ms half-frame at 120 kHz SCS).

[23] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023, doi: 10.1561/2200000101.

[24] Y. Romano, M. Sesia, and E. J. Candès, "Classification with valid and adaptive coverage," in *Proc. NeurIPS*, vol. 33, 2020.

[25] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, vol. 30, 2017, pp. 3146–3154.